



Fine-grained Hate Speech Detection in Arabic Text with Linguistic Analysis using Transformers

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Abstract

Hate speech detection in Arabic social media text is an important and challenging natural language processing task. Arabic online content often includes dialectal expressions, spelling variations, informal writing, repeated letters, hashtags, usernames, and noisy text structures. These characteristics make automatic classification more difficult, especially when harmful content appears implicitly or when hate speech examples are limited in the dataset.

This paper presents a transformer-based approach for Arabic hate speech detection using the L-HSAB dataset. The dataset contains Arabic tweets labeled into three classes: normal, abusive, and hate. Instead of relying only on traditional TF-IDF and classical machine learning classifiers, this study focuses on fine-tuning Arabic transformer models that are more suitable for contextual understanding. Three transformer models were evaluated: AraBERT, MARBERT, and Arabic-BERT.

The experimental results show that MARBERT achieved the best overall performance, with 87.01% accuracy and 79.49% macro F1-score. Arabic-BERT achieved 80.17% accuracy and 71.69% macro F1-score, while AraBERT achieved 81.62% accuracy and 71.32% macro F1-score. MARBERT performed best because it was pre-trained on large-scale Arabic social media text, making it more suitable for dialectal and informal Arabic tweets. The results also show that the hate class remains more difficult to classify than the normal and abusive classes. Overall, the evaluated transformer models achieved strong performance in Arabic hate speech classification, particularly when the pre-training data matched the dialectal and informal nature of the dataset.

1. Introduction

Social media platforms have become widely used in Arabic-speaking communities for communication, news sharing, political discussion, and daily interaction. However, these platforms also contain harmful content such as hate speech, abusive language, and offensive expressions. Hate speech can negatively affect individuals and communities, increase social tension, and create unsafe online environments. Therefore, automatic detection of harmful Arabic text has become an important research problem in natural language processing [1].

Arabic hate speech detection is challenging for several reasons. First, Arabic has many dialects and informal writing styles. Users often write in dialectal Arabic rather than Modern Standard Arabic. Second, social media posts are usually short, noisy, and full of spelling variations, repeated letters, emojis, mentions, hashtags, and informal expressions. Third, hate speech may be explicit or implicit. Some hateful posts contain direct offensive words, while others express harmful meaning through context, sarcasm, or indirect references.

Traditional machine learning methods such as Logistic Regression, Support Vector Machines, and Naive Bayes with TF-IDF features can provide useful baselines, but they are limited because they mainly depend on surface-level word or character patterns. These methods may fail when the

meaning depends on context, target group, sarcasm, dialect, or implicit harmful expressions. For this reason, transformer-based models have become more suitable for modern Arabic NLP tasks because they learn contextual representations from large amounts of text [3], [4].

This study focuses on Arabic hate speech classification using the L-HSAB dataset. Each tweet is classified into one of three classes: normal, abusive, or hate. This study evaluates three Arabic transformer models: AraBERT, MARBERT, and Arabic-BERT. The main goal is to compare how well these transformer models handle Arabic social media text and to identify the best model for fine-grained hate speech detection.

The main contributions of this paper are:

1. Fine-tuning and comparing three Arabic transformer models for Arabic hate speech detection: AraBERT, MARBERT, and Arabic-BERT.
2. Evaluating the models using accuracy, macro precision, macro recall, and macro F1-score on the L-HSAB dataset.
3. Analyzing the class-level performance of the best model across normal, abusive, and hate classes.
4. Providing linguistic and error analysis to explain why some Arabic tweets are easier or harder to classify.

2. Related Work

Arabic hate speech detection has been studied using traditional machine learning, deep learning, and transformer-based approaches. Early studies commonly used traditional classifiers such as Support Vector Machines, Logistic Regression, Random Forest, and Naive Bayes. These models usually depend on manually designed features such as word n-grams, character n-grams, bag-of-words, or TF-IDF representations. Traditional methods are fast, simple, and easier to interpret, but they often struggle with context, sarcasm, dialectal expressions, and implicit hate speech.

Several early Arabic studies relied on traditional machine learning approaches and handcrafted features for detecting offensive and antisocial online behavior [1], [15].

Shared tasks have played an important role in establishing common benchmarks for Arabic harmful-content detection. OSACT4 included tasks for offensive language and hate speech detection, while OSACT5 extended the benchmark to include more fine-grained hate speech categories. These shared tasks enabled researchers to compare different systems using common Arabic datasets and evaluation settings [11], [14], [16].

Deep learning models such as CNN, RNN, LSTM, and GRU improved text classification by learning features automatically from data. Some systems used word embeddings and ensemble methods to improve performance. However, these models still require sufficient labeled data and may struggle with long-range context, rare dialectal expressions, and implicit meaning.

Transformer-based and ensemble approaches later achieved stronger performance in Arabic hate speech detection benchmarks and shared tasks [9], [10].

AraBERT was introduced as an Arabic-specific transformer model designed for Arabic language understanding tasks [13]. Later, ARBERT and MARBERT extended transformer pretraining to larger Arabic corpora, with MARBERT focusing heavily on social media and dialectal Arabic text [12].

Recent studies have demonstrated the effectiveness of BERT-based architectures for Arabic hate speech detection and fine-grained offensive language classification [5], [6], [10]. Recent research has also focused on identifying hate speech targets and fine-grained categories using dedicated resources such as AraTar [8].

Despite this progress, several challenges remain. Arabic hate speech datasets are frequently imbalanced, with considerably fewer hate examples than normal or offensive examples. Moreover, many studies emphasize overall performance without providing detailed class-level or linguistic error analysis. Therefore, this study compares three Arabic-specific transformer models under the same experimental setting and analyzes their performance across the normal, abusive, and hate classes. It also investigates the linguistic characteristics associated with the errors produced by the best-performing model.

3. Methodology

This section describes the dataset, preprocessing steps, transformer models, and training approach used in this study. The methodology focuses on transformer-based models because contextual representations are better suited to dialectal, informal, and context-dependent Arabic social media text than sparse feature-based representations.

3.1 Dataset

This study uses the L-HSAB Arabic hate speech dataset [2]. The original dataset contains 5,846 Arabic tweets distributed as follows: 3,650 normal tweets, 1,728 abusive tweets, and 468 hate tweets. After removing one empty or invalid record during data cleaning, 5,845 tweets remained for the experimental data splits. This distribution demonstrates a clear class imbalance, as the hate class contains substantially fewer examples than the normal and abusive classes.

In the experimental setup, the cleaned training split contained 3,973 tweets, the validation split contained 702 tweets, and the final test split contained 1,170 tweets. The same label set was used in all splits: normal, abusive, and hate. The normal class represents neutral or non-harmful content. The abusive class represents offensive or insulting language that may not necessarily target a protected group. The hate class represents harmful content that targets individuals or groups based on identity, affiliation, or other shared characteristics.

3.2 Text Preprocessing

Arabic social media text is usually noisy and informal. It may include links, usernames, hashtags, spelling variations, repeated letters, and dialectal expressions. Therefore, light preprocessing was applied before tokenization. The original labels and class distribution were not modified.

The preprocessing steps included removing URLs and usernames, removing the hashtag symbol while keeping the hashtag word, normalizing Arabic letters, reducing repeated letters, removing unnecessary non-Arabic symbols, and removing extra spaces. Arabic normalization included converting different forms of Alef, such as ا, أ, and آ, to ا; converting ع to ع; converting و to و; converting ه to ه; and converting ة to ة.

Aggressive preprocessing was avoided because dialectal spelling and informal writing patterns may contain useful information for distinguishing harmful from non-harmful content. The cleaning process therefore focused on reducing excessive noise while preserving the main lexical and contextual characteristics of each tweet.

3.3 Transformer Tokenization

Unlike TF-IDF methods, transformer models do not represent text using manually designed word or character features. Instead, each transformer uses its own tokenizer to split the input text into subword tokens. Subword tokenization is useful for Arabic because Arabic has rich morphology, dialectal variations, and many spelling forms. It also helps reduce the out-of-vocabulary problem because rare words can be represented as smaller subword units.

Each tweet was tokenized using the tokenizer associated with its transformer checkpoint. Padding and truncation were applied to create fixed-length input sequences. The tokenized inputs were then passed to the transformer model for sequence classification.

3.4 Transformer Models

Three Arabic transformer models were evaluated in this study:

1. AraBERT is an Arabic-specific transformer model designed for Arabic language understanding tasks [13].
2. MARBERT is an Arabic transformer model pretrained on large-scale Arabic social media content. Its pre-training data make it particularly relevant to dialectal and informal Arabic text [12].
3. Arabic-BERT is a general-purpose Arabic BERT model used in this study as an additional transformer baseline for Arabic text classification.

For each model, a sequence classification head was added on top of the transformer encoder. The classification head outputs one of the three labels: normal, abusive, or hate.

3.5 Proposed Approach

The proposed approach is a transformer-only classification pipeline. The system can be summarized as follows:

- Input: cleaned Arabic tweet.
- Tokenization: model-specific subword tokenizer.
- Encoder: Arabic transformer model.
- Classification head: fully connected layer for three classes.
- Output: one of three labels: normal, abusive, or hate.

Class weights were used in the loss function to reduce the effect of class imbalance. Since the hate class has fewer examples than the normal and abusive classes, weighted cross-entropy gives more importance to minority classes during training. This helps the model pay more attention to hate speech examples instead of becoming biased toward the majority normal class.

4. Experiments

4.1 Experimental Setup

The task is Arabic hate speech classification using three classes: normal, abusive, and hate. The input is Arabic social media text, and the output is the predicted class label. The models were trained and evaluated using a GPU environment. PyTorch and the Hugging Face Transformers library were used to fine-tune and evaluate the three models [3].

The following models were evaluated:

1. AraBERT using the checkpoint aubmindlab/bert-base-arabertv02.
2. MARBERT using the checkpoint UBC-NLP/MARBERT.
3. Arabic-BERT using the checkpoint asafaya/bert-base-arabic.

The three models were trained using the same hyperparameter configuration to ensure a fair comparison. The experimental settings are summarized in Table 1.

Parameter	Value
Number of epochs	3
Learning rate	2×10^{-5}
Training batch size	16
Evaluation batch size	16
Maximum sequence length	128 tokens
Optimizer	AdamW (Hugging Face Trainer default)
Weight decay	0.01
Warmup steps	0
Validation size	15% of the original training set

Validation splitting method	Stratified split
Random seed	42
Evaluation strategy	Once after each epoch
Padding strategy	Maximum-length padding
Truncation	Enabled
Loss function	Weighted cross-entropy
Abusive class weight	1.1281
Hate class weight	4.1646
Normal class weight	0.5338
Checkpoint saving	Disabled
Best-checkpoint loading	Disabled
Final model used for testing	Model obtained after the third epoch
Hardware	NVIDIA Tesla T4 GPU
PyTorch version	2.11.0+cu128
Transformers version	4.44.2

Table 1. Fine-tuning and experimental settings

Each model was fine-tuned for three epochs. The same preprocessing, data splits, evaluation metrics, and random seed were used to ensure a fair comparison between the models.

4.2 Evaluation Metrics

The models were evaluated using Accuracy, Macro Precision, Macro Recall, and Macro F1-score. Macro averages were used because the dataset is imbalanced. Macro F1-score is especially important because it gives equal importance to all classes, including the minority hate class.

Accuracy alone may be misleading in imbalanced datasets. A model can achieve high accuracy by performing well on the majority class while performing poorly on minority classes. Therefore, macro F1-score was used as the main comparison metric.

5. Results

The final transformer-only results are shown in Table 2.

Model	Checkpoint	Accuracy	Macro Precision	Macro Recall	Macro F1-score
MARBERT	UBC-NLP/MARBERT	87.01%	80.49%	78.74%	79.49%
Arabic-BERT	asafaya/bert-base-arabic	80.17%	70.51%	73.16%	71.69%
AraBERT	aubmindlab/bert-base-arabertv02	81.62%	70.37%	72.60%	71.32%

Table 2. Comparison of Arabic transformer models

The best model was MARBERT, which achieved 87.01% accuracy and 79.49% macro F1-score. Arabic-BERT achieved 80.17% accuracy and 71.69% macro F1-score, while AraBERT achieved 81.62% accuracy and 71.32% macro F1-score. Although AraBERT achieved slightly higher accuracy than Arabic-BERT, Arabic-BERT achieved a slightly better macro F1-score. Since the dataset is imbalanced, macro F1-score is more important than accuracy because it gives equal importance to all classes, including the minority hate class.

The improvement of MARBERT over Arabic-BERT was 6.84 percentage points in accuracy and 7.79 percentage points in macro F1-score. The improvement of MARBERT over AraBERT was 5.39 percentage points in accuracy and 8.16 percentage points in macro F1-score. These results show that MARBERT was the best-performing model among the three evaluated models on the current test split.

Comparison	Accuracy Difference	Macro F1 Difference
MARBERT vs. Arabic-BERT	+6.84%	+7.79%
MARBERT vs. AraBERT	+5.39%	+8.16%

Table 3. Improvement of the best model over other transformer models

5.1 Detailed Results of the Best Model

The detailed classification report of the best model, MARBERT, is shown in Table 4.

Class	Precision	Recall	F1-score	Support
abusive	0.79	0.84	0.81	346
hate	0.70	0.61	0.65	94
normal	0.93	0.92	0.93	730
accuracy			0.87	1170
macro avg	0.80	0.79	0.79	1170
weighted avg	0.87	0.87	0.87	1170

Table 4. Classification report for MARBERT on the test set

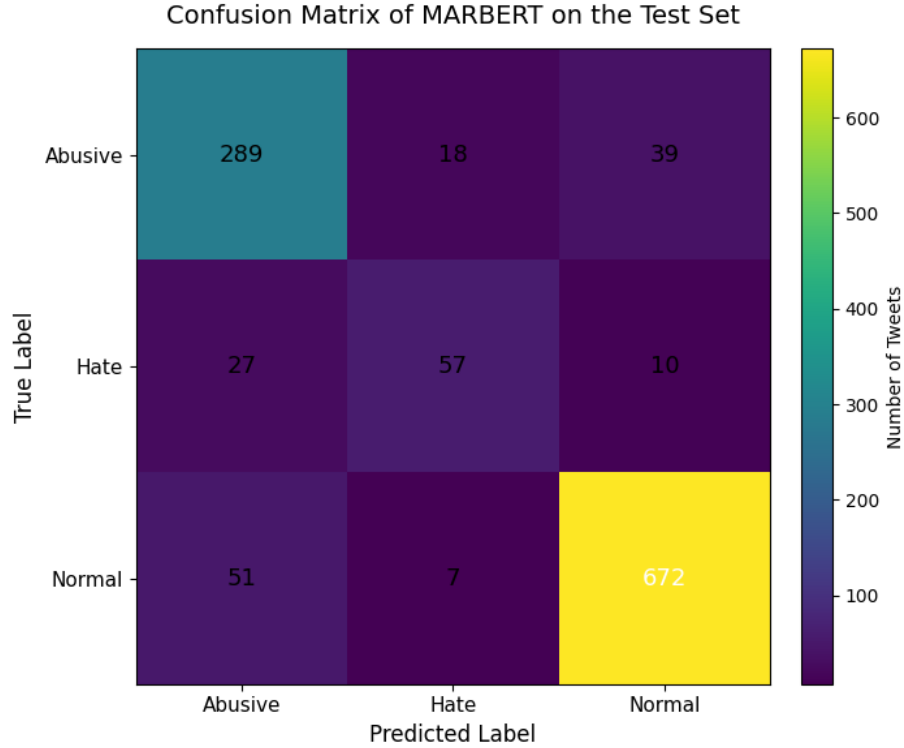


Figure 1. Confusion matrix of MARBERT on the test set.

MARBERT performed best on the normal class, with an F1-score of 0.93. It also achieved strong performance on the abusive class, with an F1-score of 0.81. The hate class remained the most difficult class, but MARBERT still achieved a hate F1-score of 0.65. This is an important improvement because the hate class is the smallest class in the dataset and is usually harder to detect than normal and abusive content.

As shown in Figure 1, the largest confusion occurred between the abusive and normal classes. Among the abusive tweets, 39 examples were incorrectly classified as normal, while 51 normal tweets were classified as abusive. The hate class remained the most difficult class to identify: 27 of the 94 hate tweets were classified as abusive, and 10 were classified as normal. These results indicate that some hate tweets share offensive linguistic characteristics with abusive content, while other harmful expressions may appear insufficiently explicit and are therefore classified as normal.

5.2 Learning Behavior

During training, MARBERT improved steadily across epochs. Its macro F1-score increased from 69.62% after the first epoch to 74.18% after the second epoch and 79.01% on the validation set after the third epoch. The final test macro F1-score was 79.49%. No decrease in validation performance was observed across the three training epochs. However,

experiments using additional epochs and multiple random seeds would be required to evaluate overfitting and training stability more reliably.

Epoch	Validation Accuracy	Validation Macro Precision	Validation Macro Recall	Validation Macro F1-score
1	79.20%	68.42%	76.85%	69.62%
2	82.05%	71.66%	78.91%	74.18%
3	87.18%	78.09%	80.09%	79.01%

Table 5. MARBERT validation performance across epochs

6. Linguistic and Error Analysis

The results show that Arabic hate speech detection is not only a classification problem but also a linguistic problem. Arabic tweets often contain dialectal words, informal spelling, repeated letters, hashtags, short context, and indirect expressions. These features make the boundary between abusive and hate speech difficult.

To conduct the qualitative error analysis, a sample of 20 misclassified test examples generated by the error-analysis code was manually reviewed and grouped according to their main linguistic and contextual characteristics.

The normal class achieved the highest performance because normal tweets usually contain fewer harmful expressions and are more frequent in the dataset. The abusive class also achieved strong performance because abusive tweets often include direct insults or negative expressions that the model can learn from context and lexical patterns.

The hate class remained the most challenging. Hate speech examples are fewer than normal and abusive examples. In addition, hate speech may be implicit, target-specific, or dependent on social and political context. Some hate tweets may not contain obvious offensive words, while some abusive tweets may contain strong insults without being hate speech. This creates confusion between the abusive and hate classes.

MARBERT produced 152 incorrect predictions among the 1,170 test examples. The reviewed errors included cases involving short context, dialectal expressions, implicit targets, political references, and overlap between abusive and hateful meanings. These cases illustrate that the distinction between abusive and hate speech cannot always be determined from offensive vocabulary alone, because the target and surrounding context are also important.

Anonymized example	True label	Predicted label	Explanation
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يريد اعمار سوريا والزباله تطمره في بيروت	normal	abusive	Negative vocabulary and sarcastic meaning
الروس المفروض ينشهبوا بالوقاحه مو بالفودكا	hate	abusive	Group-targeted insult confused with general abuse
ما حارقلكم قلبكم غير لبنان يا كلاب	abusive	hate	Plural insult interpreted as group-targeted hate
كذبك يا ايوب	normal	abusive	Very short context and negative lexical cue

Table 6. Examples of MARBERT classification errors.

A likely explanation for MARBERT’s stronger performance is that it was pretrained on large-scale Arabic social media text [12]. This makes it more familiar with dialectal Arabic, informal expressions, noisy spelling, hashtags, and short tweet-like structures. Since L-HSAB is a tweet-based dataset, MARBERT has a clear advantage over models trained mainly on more formal or general Arabic text.

7. Discussion

The experimental results show that transformer-based models are highly effective for Arabic hate speech detection. MARBERT achieved the best performance with 87.01% accuracy and 79.49% macro F1-score. This result is important because macro F1-score gives equal weight to normal, abusive, and hate classes, which is more suitable for an imbalanced dataset.

A likely explanation for MARBERT’s strong performance is the similarity between its social-media pre-training data and the tweet-based L-HSAB dataset. MARBERT was trained on Arabic tweets and social media text, while the L-HSAB dataset also consists of Arabic tweets. This similarity helps the model understand dialectal expressions, informal spellings, and offensive language patterns more effectively.

AraBERT and Arabic-BERT also achieved useful results, but both performed lower than MARBERT in macro F1-score. This suggests that not all Arabic transformer models perform equally well on noisy social media hate speech data. Models trained on social media Arabic are more suitable for tweet-based classification tasks than models trained mainly on formal or general Arabic text.

The hate class still achieved lower performance than the normal and abusive classes. This can be explained by class imbalance and contextual complexity. Hate speech is often harder to identify because it may require knowing the target, the social context, or the implied

meaning. Some examples may look abusive on the surface but are not hate speech, while others may be hateful without using explicit hate words.

One limitation of this experiment is that duplicate and near-duplicate tweets were not fully controlled across the training and test splits. Such overlap may inflate the reported performance. Future experiments should therefore apply deduplication before data splitting and retrain all evaluated models using the revised splits.

Overall, the results support the use of transformer-based models for Arabic hate speech detection, especially models trained on social media Arabic. The findings also show the importance of evaluating class-level performance rather than relying only on accuracy.

8. Conclusion

This paper presented a transformer-based approach for Arabic hate speech detection using the L-HSAB dataset. Three Arabic transformer models were fine-tuned and evaluated: AraBERT, MARBERT, and Arabic-BERT. The task was to classify Arabic tweets into three classes: normal, abusive, and hate.

The results showed that MARBERT achieved the best performance, with 87.01% accuracy and 79.49% macro F1-score. Arabic-BERT achieved 80.17% accuracy and 71.69% macro F1-score, while AraBERT achieved 81.62% accuracy and 71.32% macro F1-score. These results show that MARBERT achieved the strongest performance among the evaluated models on the current L-HSAB split.

The class-level results showed that MARBERT performed very well on the normal and abusive classes, while the hate class remained more difficult. The hate class difficulty is mainly due to class imbalance, implicit meaning, target-specific language, and contextual dependency. Future work should evaluate the models on deduplicated data splits, conduct multiple training runs using different random seeds, investigate data augmentation for the minority hate class, and extend the classification framework to identify hate targets, emotions, sarcasm, and implicit hateful meaning.

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